

Faculty of Medicine Annaba
1^{er} year of medicine physiology
course

Ligand-receptor interactions

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PLAN:

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- VI. THE SECOND MESSENGERS*

I/ Introduction

Cellular communications by messengers chemicals are essential to organisms multicellular for the regulation of Functioning organs and the maintenance of homeostasis.

*Chemical messengers intercellular or intracellular:
ligands*

*Binding of the ligand to the receptor results in a series of events leading to the transformation of the chemical messenger signal in biological response
:signal transduction.*

II/ MODES OF INTERCELLULAR COMMUNICATION:

There are several modes of intercellular communication, we distinguish:

a. Direct communication:*involves physical contact between cells by:*

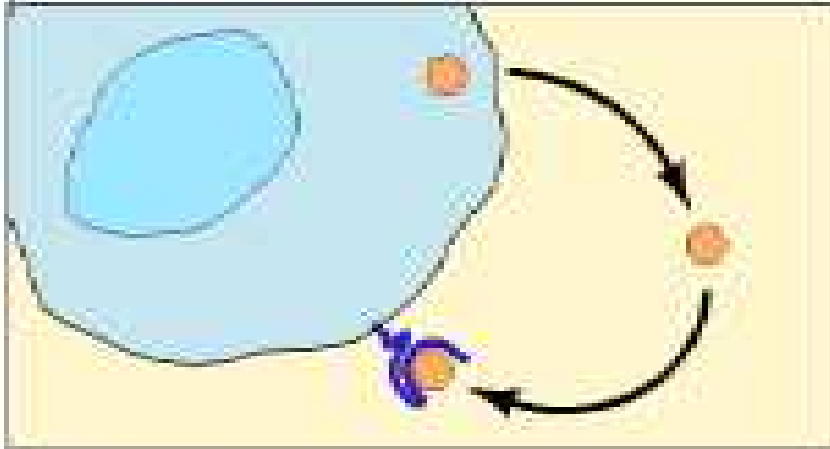
- **Communicating junctions (or gaps junction) :** *of the transmembrane proteins that form channels ionic, which allow the direct passage of certain small molecules and ions from one cell to another.*
- **Nanotubules:** *allow the passage of molecules signal and even organelles (such as mitochondria)*
- **Via surface markers:** *concerns the especially the immune system, where there are interactions between cells through specific markers at the level of cell surfaces.*
- **Adhesions**

b) Indirect communication: by signaling molecules called Sometimes

ligands On distinguishes 04 types of signaling:

- **Neurocrine (or synaptic):** relating to nerve transmission, it is based on a signal molecule called: neurotransmitter or neurohormone, secreted by cells nerves and acting on receptors postsynaptic of the cells located in their immediate vicinity at the synaptic junction. neuro-neuronal, effector neuron,
- **Endocrine:** involves a signal molecule, called a hormone, secreted by a endocrine cell and transported in the circulation to act remotely on of the target cells.
- **Paracrine:** is based on a signal molecule secreted by cells and acting on other cells located in their immediate vicinity such as neurotransmitters and local chemical mediators. The action being limited in time and space.
- **Autocrine:** is defined by cells responding to signal molecules that they produce themselves. The action being limited in time and space.

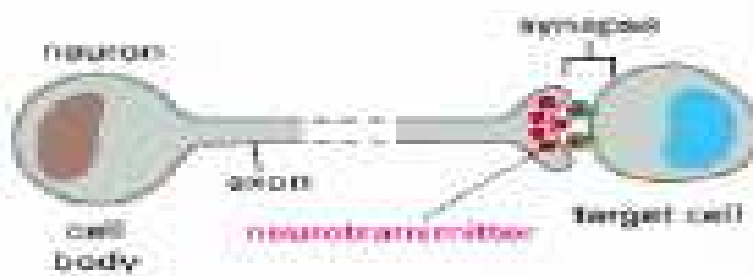
(A) AUTOCRINE



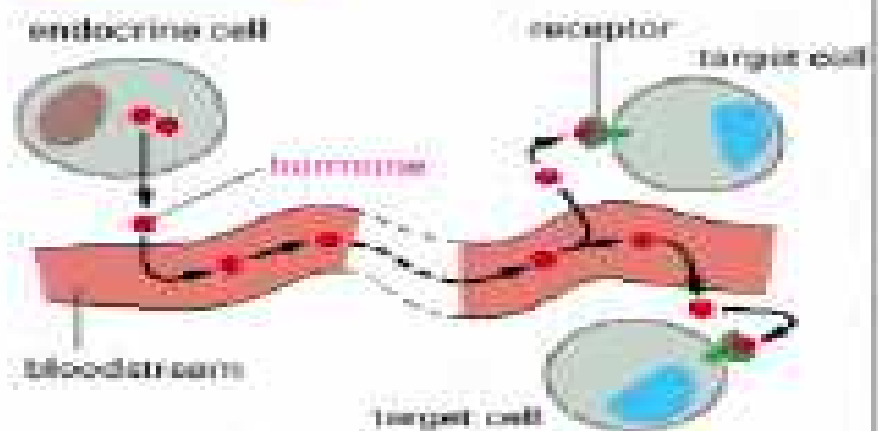
(B) PARACRINE



(D) SYNAPTIC



(C) ENDOCRINE



III/ THE PHASES OF INTERCELLULAR COMMUNICATION

Intercellular communication involves three phases: signal reception, signal transduction and the cellular response phase:

1. The Reception phase:*which consists of the detection and recognition of ligand and binding with the specific receptor.*

2. The transduction phase:*the process by which a signal is converted into a particular cellular response, the transduction phase requires the most of the time a cascade of interactions between several molecules intermediaries (transduction pathway).*

3. The cellular response phase:*The ligand-receptor interaction generates a cellular response, that is, changes in behavior cellular, which are of three orders:*

- Changes in membrane permeability to ions and water,*
- Changes in enzymatic activities on the surface and inside the cell,*
- Changes in activities transcriptional.*

IV/ LIGANDS AND RECEPTORS

- LIGANDS:

- *Ligands* = signaling molecules that bind to receptors to induce a cellular response.
- *There are two main classes:*
 - *Lipid-soluble ligands*
 - *Hydrophilic ligands*

- *Lipid-soluble ligands*: characterized by their ability to diffuse across the membrane cellular and bind to their *intracellular receptors* which can be *cytosolic* Or *nuclear*. Ex: steroid Hormones cholesterol derivatives (cortisol, aldosterone, estrogen, progesterone and testosterone), the thyroid hormones derived from acid amino tyrosine secreted by the thyroid gland.

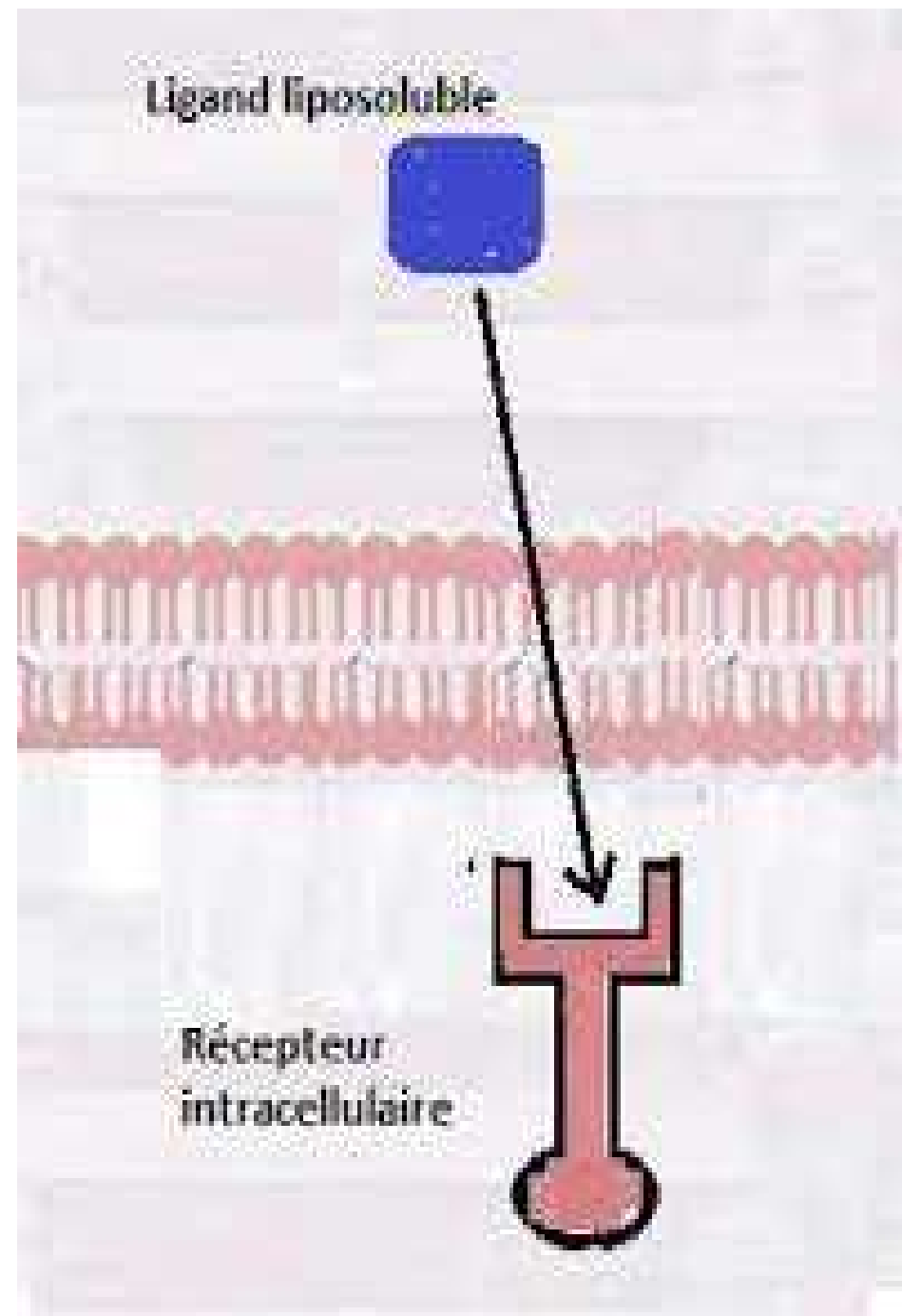
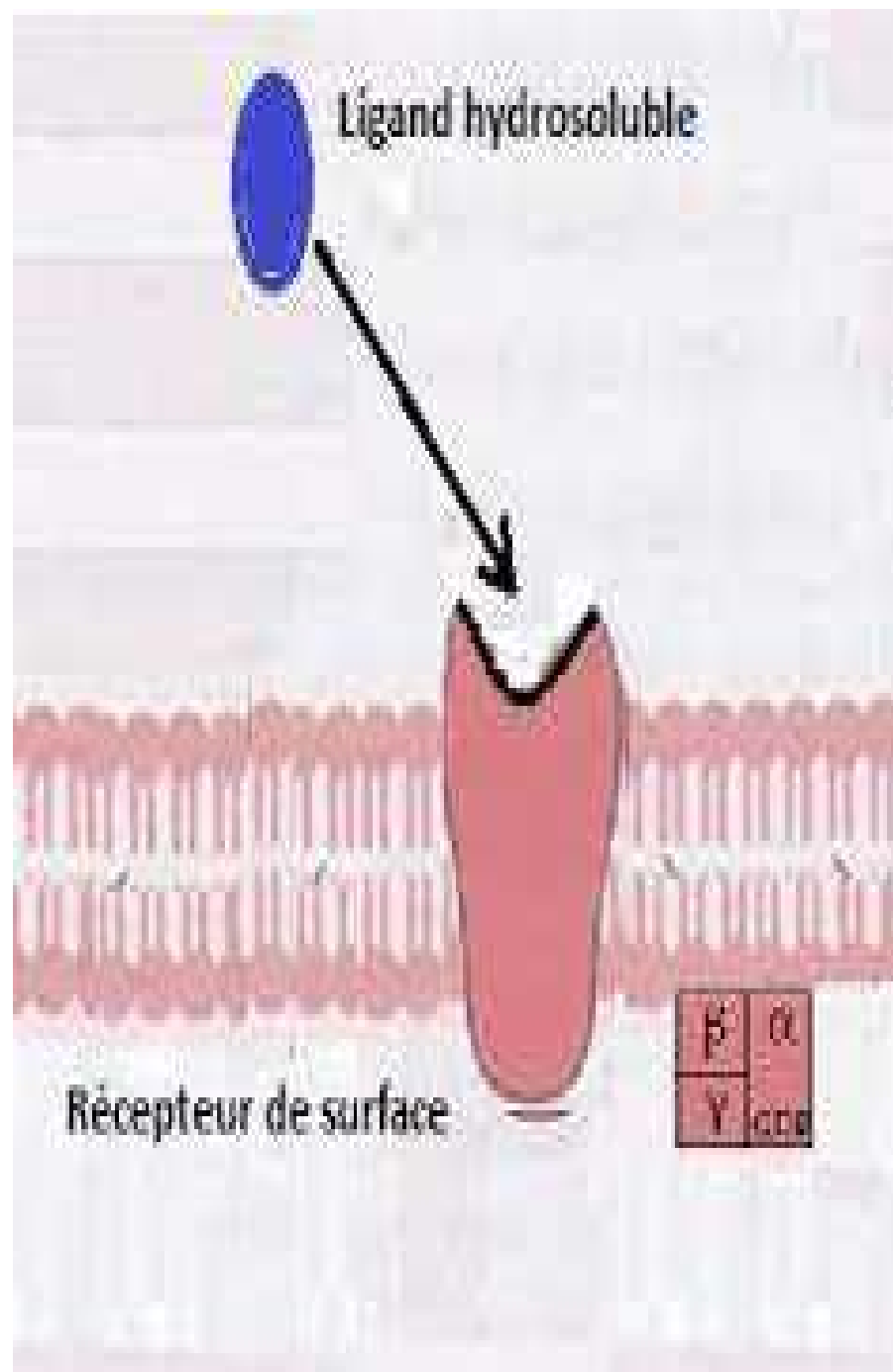
- *hydrophilic ligands*: characterized by their inability to diffuse across the membrane cellular. They bind to *membranes receptors*. Ex: Neurotransmitters (noradrenaline, acetylcholine, serotonin, etc.), peptide hormones and growth factors.

-RECEPTORS

Receptors=proteins with binding sites for ligands, located *in the cell* (cytosol or noyau) or *in the plasma membrane*

There are several types of receptors:

- intracellular receptors* represented by:
 - cytosolic receptors*: located in the cytosol
 - nuclear receptors*: located in the noyau
- membrane receptors* represented by:
 - channel- receptors*
 - Enzymes- receptors*
 - G protein-coupled receptors*



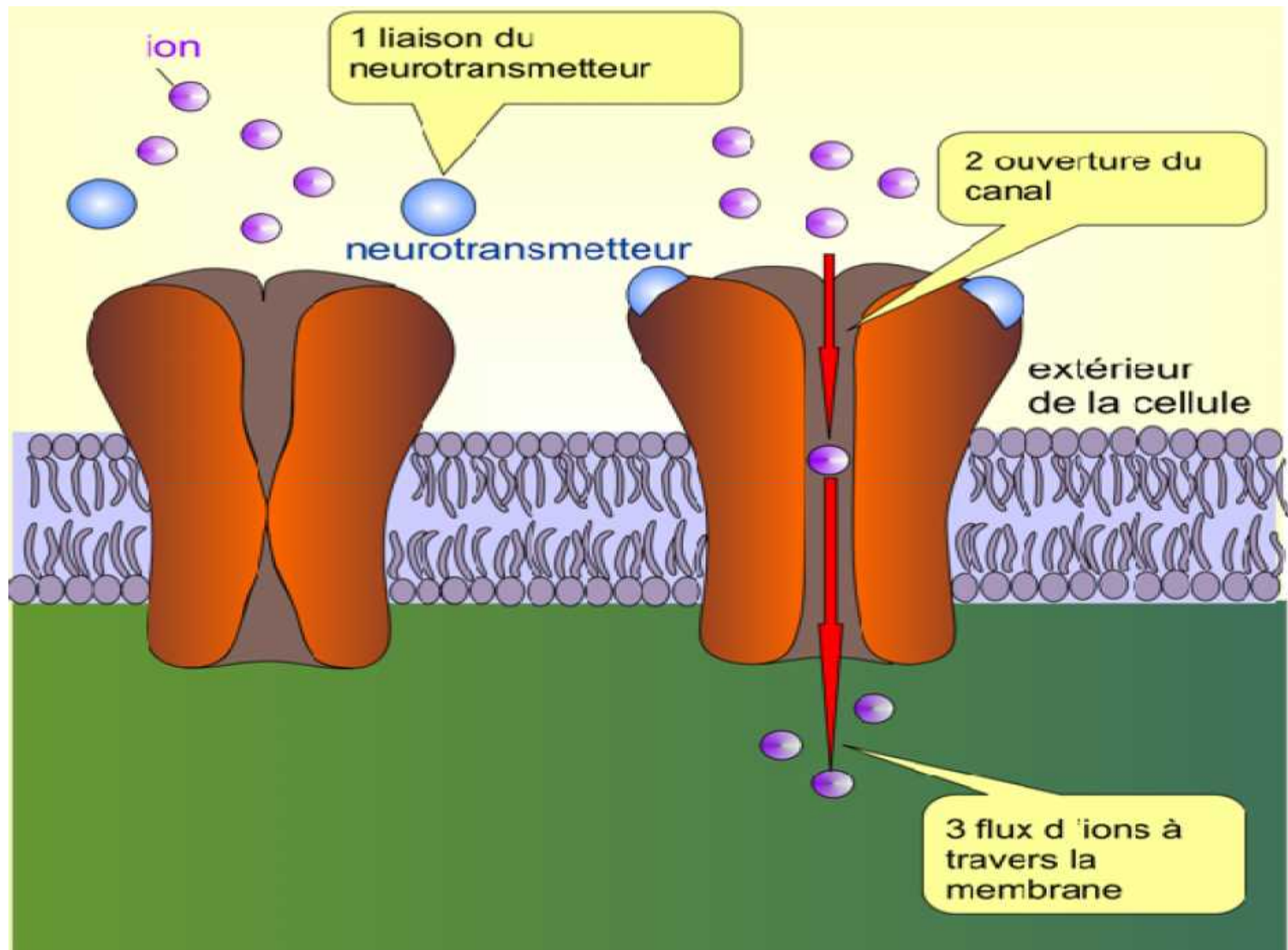
V.MECHANISMS LIGAND-RECEPTOR INTERACTIONS

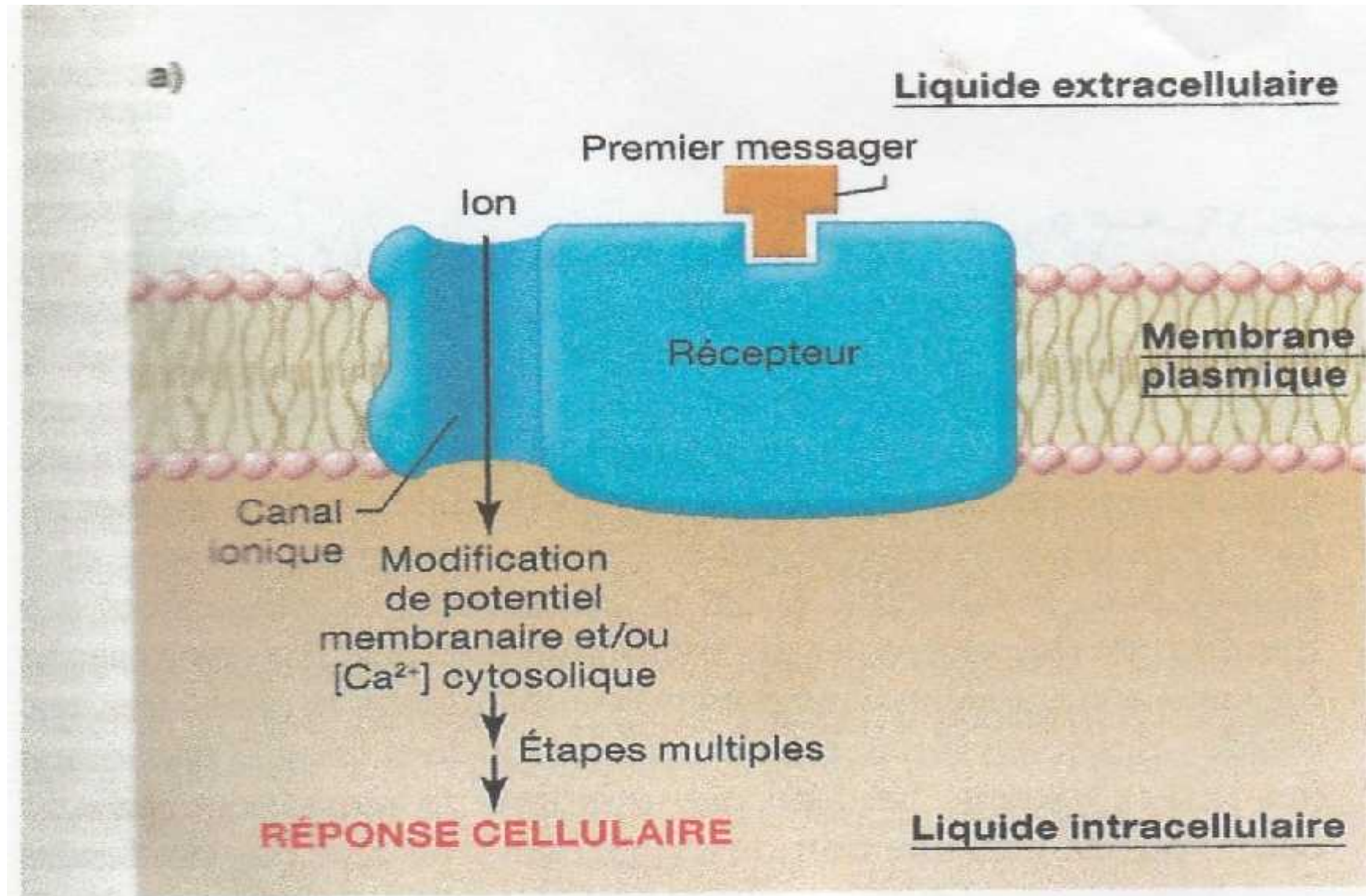
1.Surface receptors:

Channel receptors (or ionotropic receptors): This mechanism mainly concerns excitable cells such as neurons or muscle cells. These receptors are transmembrane channel proteins which have ligand binding sites.

Transduction mechanism: the binding of the ligand (neurotransmitter for example) to its receptor causes the modification of the conformation of the channel protein which becomes permeable to certain ions.

Example: nicotinic cholinergic receptor (nicotinic receptor-channel acetylcholine).





Signal transduction mechanism in which the receptor complex includes an ion channel

b) G protein-coupled receptors(RCPG):These are receptors linked to a G protein, the latter are composed of three α subunits (α_s for stimulation α_i for inhibition), β , and γ . The α subunit has a binding site with GDP/GTP.

Transduction mechanism:The message must go through 7 stages, in order to be received and correctly interpreted by the target cell.

- The ligand binds to the RCPG which is specific to it;
- The receptor modifies its three-dimensional conformation and activates the G protein
- The GDP leaves the α subunit of Protein G (PG);
- The GTP replaces GDP at the nucleotide binding site of the subunit α ;
- The α subunit is activated by binding with GTP and detaches from the dimer $\beta\gamma$ of G protein;

- The two α -GTP and β - γ complexes activate, each on its own, a effector different ;
- Hydrolysis of GTP to GDP by activity GTPase intrinsic of the below unit α , the molecule α GDP reassociates with the β - γ dimer.

The binding of the ligand (first messenger) to its receptor causes

Activation of the subunit α s, which itself activates an effector

Primary (an enzyme), which in turn stimulates the formation of

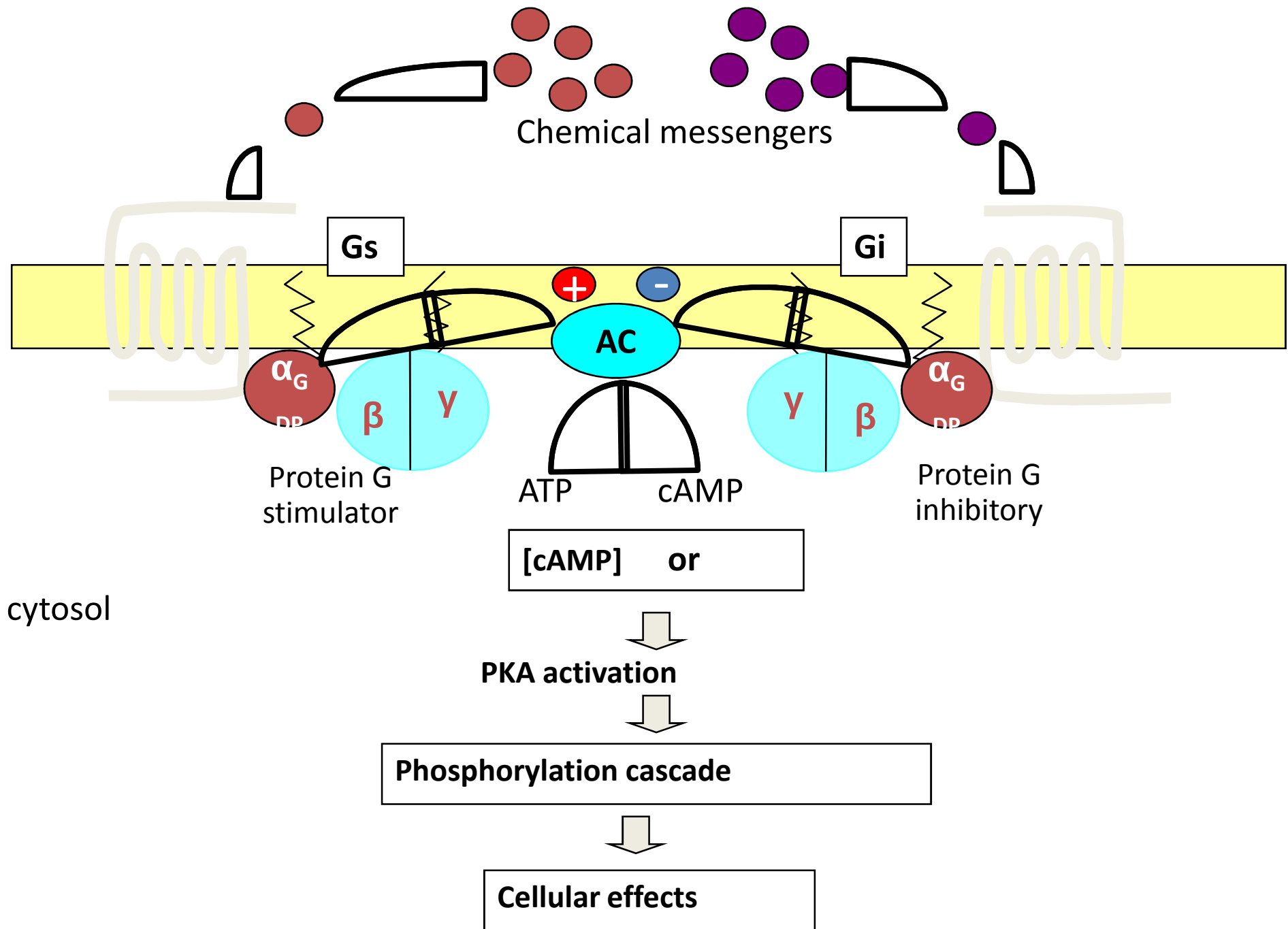
Seconds messengers, the latter can activate proteins Kinases

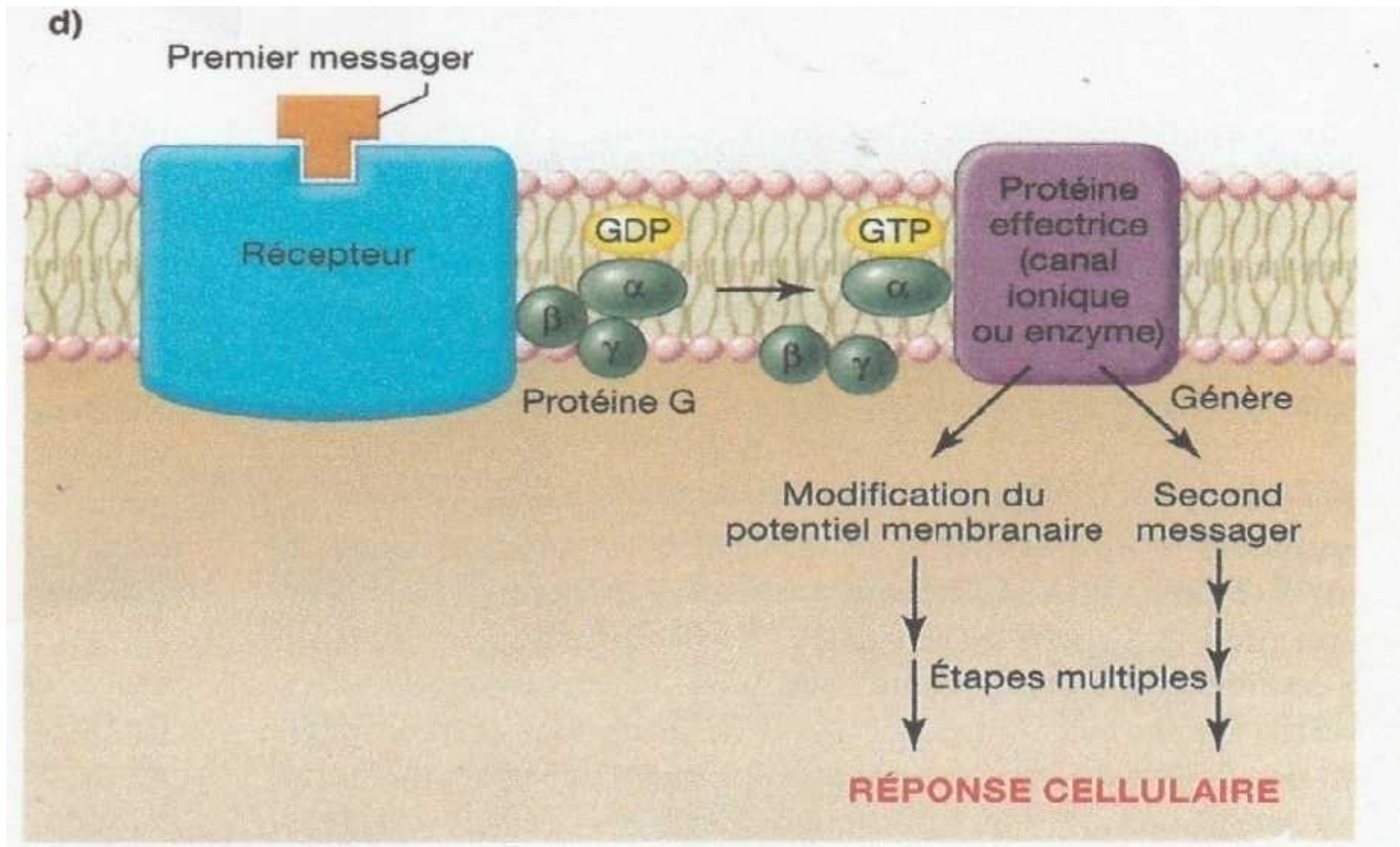
intracellular to cause several biological effects.

More than 50% of current therapeutic agents target the

RCPG.

Examples: adrenaline receptor, muscarinic receptor of acetylcholine.



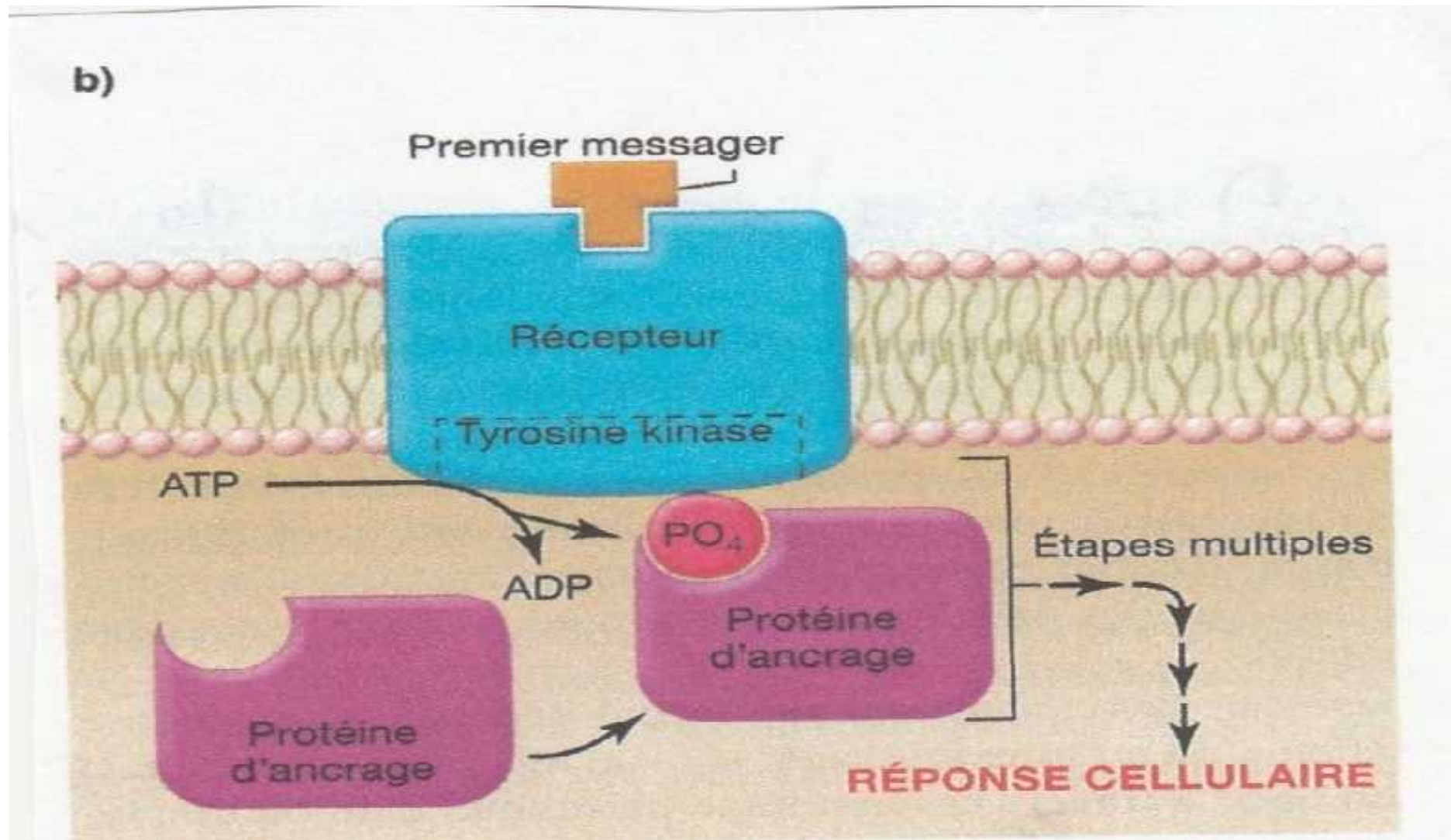


Signal transduction mechanism involving proteins

c) Enzyme receptors (tyrosine kinases receptor) :

Are enzymatic membrane receptors. Structurally, these Receptors are individual polypeptides (monomers), which have Each an extracellular binding site (which binds with a ligand) and a Domain cytoplasmic, or associated with several tyrosine kinase molecules intrinsic, or associate with cytoplasmic tyrosine kinases when it is activated.

Mechanism of transduction:*The transduction mechanism is the Even for both types, the binding of ligands to the domains Extracellular polypeptides results in the close association of two Polypeptides receptors, which forms a dimer (dimerization), there dimerization activates the tyrosine kinase of each polypeptide, which results in the phosphorylation of the latter by hydrolysis of ATP, once the tyrosine Kinase phosphorylated, they can be recognized by intermediaries Protein intracellular, thus activated Each activated protein primes A pathway (cellular response). Examples: insulin.*



Signal transduction mechanism in which receptor behaves like an enzyme

2. Intracellular receptors: located either in the cytoplasm or

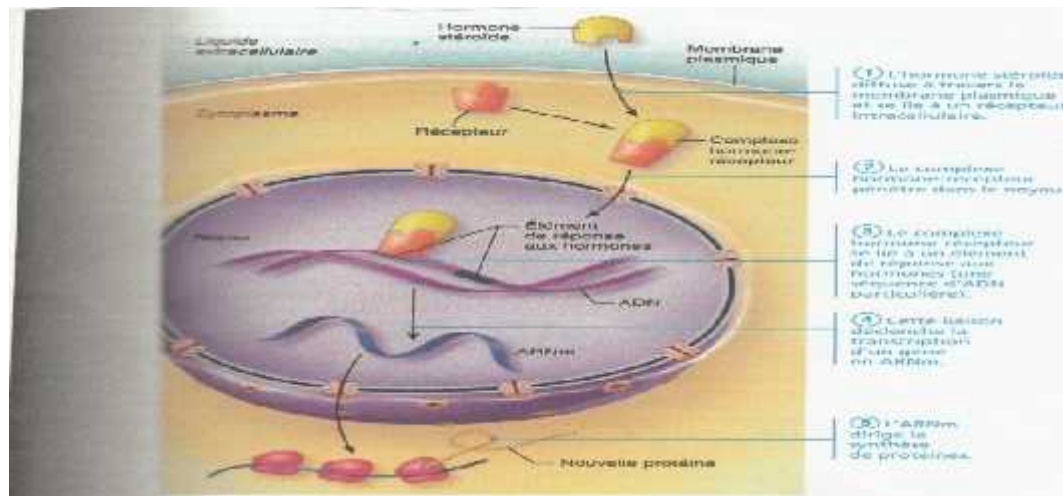
in the nucleus of target cells .

Mechanism of transduction: these are the liposoluble ligands, cross the plasma membrane of target cells, then binds to a Cytosol receptors-specific, ligand-receptor complex

Penetrates in the nucleus and attaches to specific genes, this will stimulate the Transcription of a gene in mRNA. L'm RNA is translated into a protein specific by cytoplasmic ribosomes.

Examples: steroid and thyroid hormones and the vitamin D.

**Direct activation
of a gene by a
lipid-soluble
ligand**



VI. THE SECOND MESSENGERS

Second messengers are small molecules synthesized (or released) in the cell in response to extracellular signaling molecules.

The main second messengers are:

- 1. Pathway of cAMP:** At rest the α subunit is bound to GDP (α s-GDP), when a ligand binds to the extracellular portion of the receptor, which contact activates the change in the conformation of the G protein which activates, the complex (α s-GDP) will be replaced by (α s-GTP), and the complex (α s-GTP) is released from the β and γ subunits, then activates and turns the adenylate cyclase (AC) (effector enzyme); The result of which is either the opening of an ion channel (the case of a ionotropic receptor), either the turning of second messengers (cAMP), which activates protein kinases, these can trigger several biological effects.
- NB:** in the case of complex (α i-GTP) the result will be the inhibition of the turning of second messengers (cAMP).

2. Pathway cGMP: Also plays the role of second messenger. It is produced at Leave of guanosine-triphosphate (GTP) by the guanylate-cyclase and degraded In GMP by a phosphodiesterase.

3. Pathway phospholipase C, IP3, DAG:

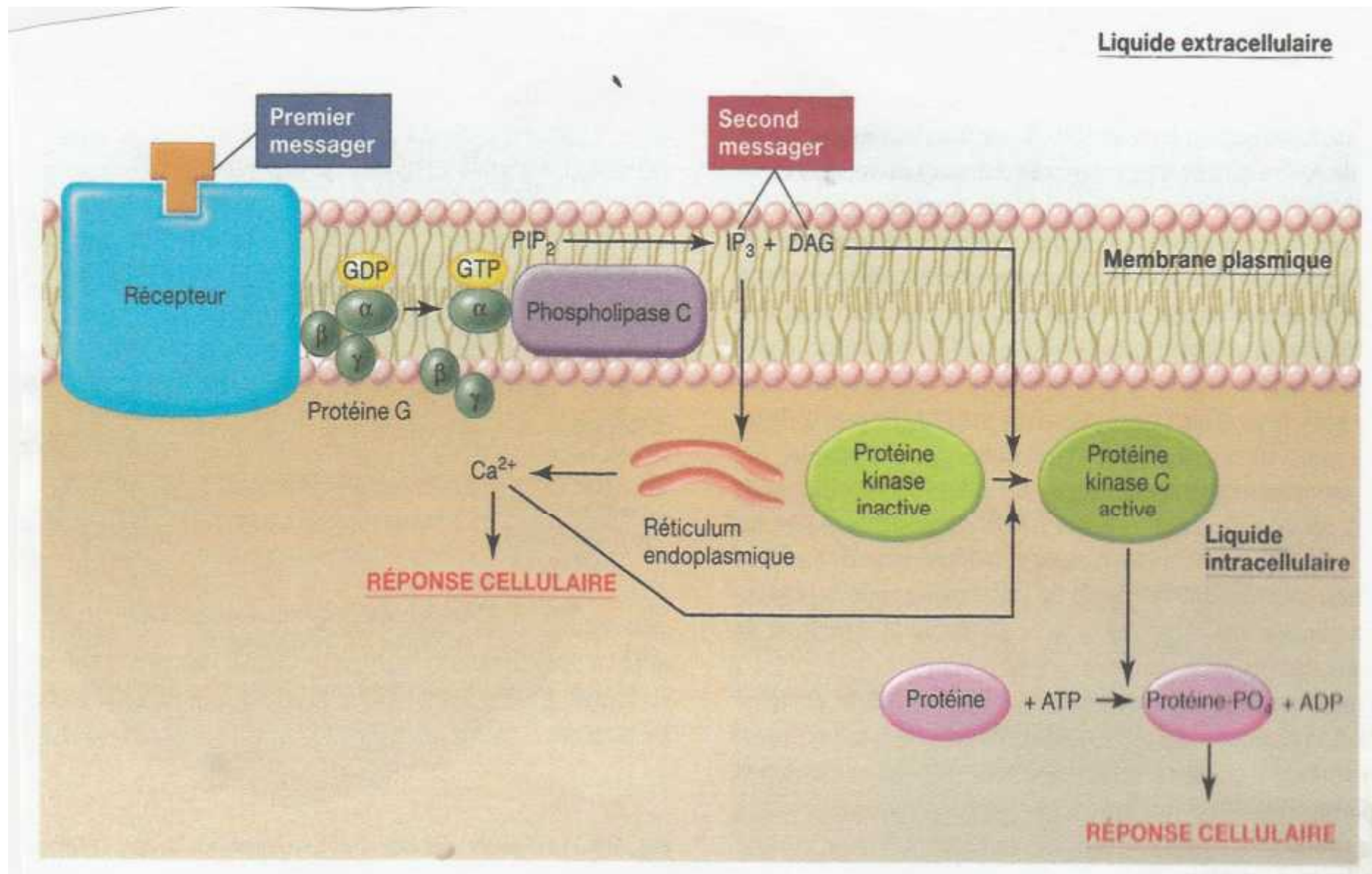
The combination of the ligand on its receptor activates via a G protein a phospholipase C which splits the phosphatidyl inositol(PI) membrane in Two second messengers: the inositoltriphosphate (IP3) and the diacylglycerol (DAG).

- DAG activates a PKC (Calcium-dependent Protein Kinase). It Phosphoryl many substrates that relay the message, in particular of the transcription factors.

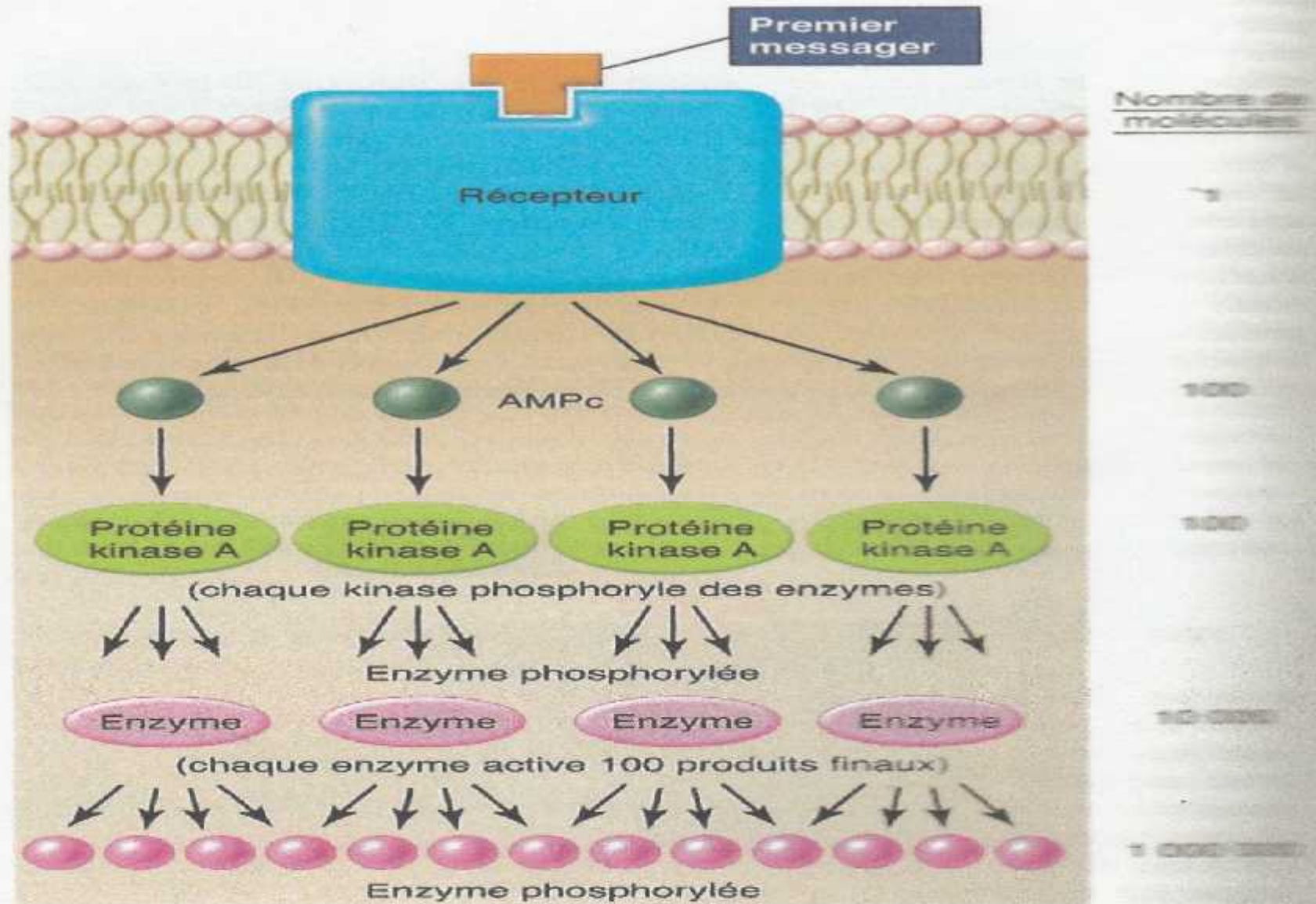
- IP3 leaves the membrane to attach itself to its receptor located on the Membranes mooth endoplasmic reticulum (**REL**).

This receptor is a Ca^{2+} channel that opens and allows the release of Ca^{2+} In the cytoplasm.

The Ca^{2+} ions bind and activate calmodulin. This then becomes Able to activate many enzymes including protein kinases Ca^{2+} /calmodulin dependent .



Mechanism by which the receptor stimulates the enzymatic catabolism of PIP_2 in IP_3 and DAG



Signal amplification in the system cAMP

THANKS